

# Grothendieck constants: current bounds and comparison

Literature before and after arXiv:2609.07000 · 10 September 2026

The classical constants compare the vector optimum of a finite bilinear form with its scalar optimum, over all matrix sizes and vector dimensions. The scalar variables are signs over  $\mathbb{R}$  and unit complex phases over  $\mathbb{C}$ . Our work gives the following bounds; neither exact constant is determined.

$$1625\pi/2917 \leq K_G^{\mathbb{R}} \leq \pi/[2(0.8818276493988)]$$
$$1.373 < K_G^{\mathbb{C}} \leq 100000000/71185999$$

## Comparison with the literature

The literature columns contain publicly stated results and their direct consequences. “Before” includes the August 2026 real-case results [1]; “after” adds Guo, Fang and Lu [4]. The “Our work” column summarizes unpublished computer-assisted proofs.

| Bound         | Before the new paper                      | After the new paper                       | Our work                                                                       |
|---------------|-------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------|
| Real lower    | $6\pi/11$ [1]<br>$\approx 1.713595992867$ | $6\pi/11$ [1]<br>$\approx 1.713595992867$ | <b><math>1625\pi/2917</math></b><br><b><math>\approx 1.750115893755</math></b> |
| Real upper    | $1.781841324234^*$ [1]                    | $1.781841324234^*$ [1]                    | <b><math>1.781296297373</math></b>                                             |
| Complex lower | $\approx 1.33807$ [3]<br>(Davie)          | $1.35584631827168$ [4]                    | <b><math>1.373</math></b>                                                      |
| Complex upper | $1.404909132736$ [2]<br>(Haagerup)        | $1.404909132736$ [2]<br>(Haagerup)        | <b><math>1.404770620695</math></b>                                             |

Entries are endpoint values. Displayed decimal upper endpoints are rounded upward; values marked  $\approx$  are approximations to the expressions shown. The exact inequalities for our work appear above. Bold entries indicate the strongest bounds in each row.

\*Theorem 6.4 of [1] states the real upper bound  $1.7818666069360661$ . Its Proposition 6.3 and Lemma 6.2, combined with Proposition 6.1, give the stronger comparison endpoint used here:

$$\pi/\{2[0.881573822049 - 0.000011328860 - 14.443/\sqrt{(10 \cdot 251^5)}]\}$$
$$= 1.78184132423347648...$$

**Size of the improvement.** Relative to these real literature bounds, the lower endpoint increases by  $0.03651990$  and the upper endpoint decreases by  $0.00054503$ , reducing the interval width by **54.31%**. The complex lower endpoint exceeds [4] by **0.01715368**. Our complex interval is **52.47%** narrower than the Davie-Haagerup interval and **35.25%** narrower than the interval after [4]. These percentages compare interval widths, not relative errors in the unknown constants.

**Other reported claims.** Li et al. [5, §3, Table 2] report internally tested real bounds  $51\pi/92 \approx 1.741535057968$  and  $1.7813319810625639$ , expressly not stated as theorems. Our endpoints are also stronger numerically, by approximately  $0.00858084$  and  $0.00003568$ , respectively.

# Methods and prior work

**Real lower bound.** The proof bounds the scalar norm of the Gaussian operator  $(13/8)P_1 - P_3$ , where  $P_j$  projects onto Hermite degree  $j$ . For signs  $f$  and  $g$ , the midpoint functions  $h = (f + g)/2$  and  $k = (f - g)/2$  satisfy  $hk = 0$  and  $h^2 + k^2 = 1$ . A circle inequality and certified one-dimensional estimates for ternary functions control every Gaussian dimension. Explicit finite approximation then gives matrix witnesses. This shares the agreement/disagreement and Gaussian-coefficient analysis of [1], with a sharper universal inequality and a direct finite-matrix conclusion.

**Real upper bound.** The construction extends Gaussian rounding and tensor preprocessing, including the coordinate-dependent framework of [1]. It uses a degree-nine polynomial threshold and a degree-eighty-one preprocessing polynomial. A lower bound for the linear correlation coefficient, minus the absolute sum of all nonlinear coefficients, gives the universal rounding guarantee. Interval enclosures cover coefficients through degree 501; a fourth-derivative bound controls the entire remaining tail. The contribution is a stronger explicit scheme and its complete coefficient estimate.

**Complex lower bound.** The operator is  $P - \lambda I - \beta C_3$ , with  $P$  the holomorphic linear projection and  $C_3$  the full real cubic chaos, complexified. The Gaussian projection framework and proposed cubic perturbation precede both constructions [4, 6]. Our separate radial weights for real and imaginary components use the exact relation  $\text{Re}(h \text{ conj}(k)) = 0$ . Pointwise maximization and a finite family of auxiliary weights bound the whole remaining scalar parameter range for one fixed operator. Exact rational certificates and finite approximation yield the bound 1.373.

**Relation to Guo-Fang-Lu.** Their common radial-weight Gram/Schur criterion estimates the scalar norm within the Gaussian multiplier family shared by the two constructions. Theorem 5.2 of [4] confines that criterion's optimum to (1.35584631827168, 1.35584697425050); this is not an upper bound for the constant or for the entire operator family. Their §5.4 identifies discarded joint phase information. Our anisotropic estimate uses this information and lies outside that restricted criterion. Rotation-character compression places our construction in their multiplier family, so the sharper scalar estimate is the substantive distinction.

**Complex upper bound.** Complex Gaussian phase rounding combines signed tensor preprocessing with nonlinear phase perturbations, including a radial damping factor. For its radial correlation expansion, the linear coefficient minus the full absolute nonlinear coefficient mass bounds the retained objective. Interval certificates control the retained terms, while analytic remainder bounds cover all scalar degrees and phase orders. This improves Haagerup's endpoint by approximately 0.000138512. The new lower-bound paper [4] does not change this upper-bound comparison.

## References

- [1] R. Saha et al. *New Lower and Upper Bounds for the Grothendieck Constant*. [arXiv:2608.11158v2](#), §§2, 6-7 (2026).
- [2] U. Haagerup. *A new upper bound for the complex Grothendieck constant*. *Israel J. Math.* 60, 199-224 (1987). [doi:10.1007/BF02790792](#).
- [3] S. Friedland, L.-H. Lim and J. Zhang. *An elementary and unified proof of Grothendieck's inequality*. *L'Enseignement Math.* 64, 327-351 (2018), §1(v). [arXiv:1711.10595](#).
- [4] S. Guo, E. X. Fang and J. Lu. *An Improved Lower Bound for the Complex Grothendieck Constant*. [arXiv:2609.07000v1](#), Theorems 1.1, 3.1, 5.2 and §5.4 (2026).
- [5] A. Li et al. *Long-Horizon AI Research for Grothendieck Constant: A Case Study in Human-AI Mathematical Collaboration*. [arXiv:2608.11195v3](#), §3, Table 2 (2026).
- [6] S. Heilman. *A lower bound for Grothendieck's constant*. [arXiv:2603.22616v1](#), §1.8 (2026).